

Image Normalization we implemented is as follows:

$$\begin{aligned}Tmp(i, j, k) &= \frac{In(i, j, k)}{255} \\ Out(i, j, k) &= \frac{Tmp(i, j, k) - \mu_k}{\sigma_k}\end{aligned}$$

And we want to turn it into shaped like:

$$w \cdot In + b = Out$$

, where w and b are the weights and bias of a convolutional layer.

We can rewrite the normalization equation as:

$$\begin{aligned}Out(i, j, k) &= \frac{\frac{In(i, j, k)}{255} - \mu_k}{\sigma_k} \\ Out(i, j, k) &= \frac{1}{\sigma_k} \left(\frac{In(i, j, k)}{255} - \mu_k \right) \\ Out(i, j, k) &= \frac{1}{255 \cdot \sigma_k} In(i, j, k) - \frac{\mu_k}{\sigma_k}\end{aligned}$$

So we can see that the weights and bias of the convolutional layer are:

$$\begin{aligned}w &= \frac{1}{255 \cdot \sigma_k} \\ b &= -\frac{\mu_k}{\sigma_k}\end{aligned}$$

Then we put value in it:

$$\begin{aligned}\mu &= (0.4914 \quad 0.4822 \quad 0.4465) \\ \sigma &= (0.2470 \quad 0.2435 \quad 0.2616)\end{aligned}$$

The final weights and bias are:

$$\begin{aligned}w &= (0.0159 \quad 0.0161 \quad 0.0150) \\ b &= (-1.9895 \quad -1.9803 \quad -1.7068)\end{aligned}$$

Since the kernel is 1×1 and the channel of output is 3, the size of weight is $1 \times 1 \times 3 \times 3$, and size of bias is $1 \times 1 \times 3$.